

Luca Moldovan

Computer Science Student • Bioinformatics & Systems Enthusiast

Bremen, Germany | moldovanluca1605@gmail.com | linkedin.com/in/luca-moldovan | github.com/moldovanluca4

EXECUTIVE SUMMARY

Computer Science student at Constructor University with a Mathematics minor. Recently wrapped up a fantastic research internship at the University of Calgary focusing on genomics representation learning and disease prediction. Dedicated to systems programming, deep learning, operating systems, and high-performance computing, actively preparing for master's degree studies and future opportunities to work, grow, and innovate.

EDUCATION

Constructor University

Bremen, Germany | Sep 2024 - Jun 2027 (Expected)

Bachelor of Science in Computer Science, Minor in Mathematics

EXPERIENCE

Research Intern — Computational Genomics & Deep Learning

Jun 2026 - Aug 2026

University of Calgary (ARC HPC Cluster) • Calgary, AB, Canada

- Awarded 2026 Mitacs Globalink & DAAD-RISE Worldwide Scholarship in Dr. Long's lab; developed TL-Prot-ISM.
- Engineered In-Silico Mutagenesis (ISM) scoring for TL-Prot, predicting steady-state lung protein abundance from 196 kb DNA windows.
- Discovered first-order Taylor gradient (TISM) failure; formulated calibrated Hybrid Selection & Audit pipeline (>90% top-hit recall).
- Built 12,000-line multi-omic annotation suite (VEP, SpliceAI, MEME/FIMO, cCREs) and orchestrated Slurm GPU jobs via PyArrow Parquet.

Teaching Assistant — Elements of Calculus

Feb 2026 - Jun 2026

Constructor University • Bremen, Germany

- Mentored undergraduate students through core mathematical concepts, multivariable calculus, and formal proofs.
- Conducted tutorial sessions, led weekly exercise discussions, and graded assignments with comprehensive feedback.

Software Development Intern

Jun 2025 - Jul 2025

Living Mainframe • Boeblingen, Germany

- Collaborated on enterprise mainframe solutions, z/OS operating environments, automation scripts, and tooling.
- Gained hands-on experience in mission-critical software architectures and low-level system workflows.

Former IBM Z Student Ambassador

Mar 2025 - Jun 2026

IBM Z Student Ambassador Program • Constructor University / Boeblingen, Germany

- Selected as guest speaker at AMC Young Talents Mainframe Convention 2025 ('Enhancing Skills with IBM Z XPLORE').
- Led campus workshops, hackathons, and technical sessions promoting mainframe architecture and enterprise computing.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Bash / Shell Scripting, SQL, HTML5 / CSS3, JavaScript

Systems & Platforms: Linux / UNIX, Git, Docker, OpenMediaVault (OMV Homelab), High-Performance Computing (ARC), z/OS

Domains & Topics: Deep Learning (PyTorch), Bioinformatics Workflows, Genomics Representation Learning, Operating Systems

HONORS, AWARDS & EXTRACURRICULAR

- Mitacs Globalink Research Internship Award (2026):** Competitive international research scholarship funded by Mitacs Canada
- DAAD-RISE Worldwide Scholarship Recipient (2026):** German Academic Exchange Service grant for research at University of Calgary
- AMC Young Talents Mainframe Convention 2025:** Invited speaker on enterprise modernization and mainframe technologies
- Erasmus+ Youth Exchange Coordinator & Member:** Led and coordinated 4 youth projects with Landessportjugend Sachsen-Anhalt